

L4 v0.2 kernel-integral mass mechanism — first F2 test of Tier 0 framework. Per Casey Sunday Lane D + Keeper synthesis. CANDIDATE derivation: $T_{190} m_\mu/m_e = (24/\pi^2)^6 = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$ structurally substrate-decomposable via Bergman kernel matrix elements on $V_{(1/2,1/2)}$ spinor radial tower; (substrate-integer numerator) / (Bergman- π -factor denominator) $^{\{Casimir\ primary\}}$ pattern. Promotion target: SEARCH-FIT $\rightarrow$ DERIVATION via explicit Bergman matrix element computation (Elie multi-week). Uses Faraut-Korányi $c_{FK} \cdot \pi^{(9/2)} = 225 + \text{Tier 0 v0.1.6 boundary unification.}$

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L4 v0.2 — kernel-integral mass mechanism (first F2 test of Tier 0)

0. Why this work + Tier 0 v0.1.6 frame

Per Casey Sunday Lane D + Keeper synthesis: derive $T_{190} m_\mu/m_e = (24/\pi^2)^6$ from Bergman kernel matrix elements on the spinor radial tower $V_{(1/2,1/2)}$ (Lepton sector ground per Tier 0). This is the first F2 test of Tier 0 v0.1.6 framework:

F2 (Tier 0 v0.1 Section 7): if the K-type level structure (C_2 eigenvalues) does NOT match observed mass ratios up to per-sector gauge subtraction, the $H_B = C_2(K)$ identification is wrong.

Per Tier 0 v0.1.6 boundary unification: mass operator lives on the Shilov boundary; bulk mass observables are holomorphic extensions of boundary matrix elements. So the F2 test computes:

$m_\lambda \propto$ matrix element of mass operator M between coherent states localized at λ 's K-type weight pattern on Shilov boundary.

Saturday's T190 work identified the STRUCTURAL integer reading $24 = N_c \cdot |W(B_2)|$ (color $\times$ Weyl-group-order). v0.2 extends this to the full closed-form reading via Bergman kernel matrix elements.

1. The substrate-decomposition reading of $(24/\pi^2)^6$

T190 closed form: $m_\mu/m_e = (24/\pi^2)^6$ at 3.4×10^{-5} precision (Tier 2 STRUCTURAL floor per Two-Tier).

v0.2 structural reading:

$$(24/\pi^2)^6 = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$$

Decomposition by substrate-primary content: - Numerator $24 = N_c \cdot |W(B_2)| = 3 \cdot 8 = 24$: substrate K-type structural integer (color $\times$ Weyl group order). Saturday-confirmed. - Denominator π^2 : Bergman volume factor $\propto$ Faraut-Korányi normalized measure $c_{FK} = 225/\pi^{(9/2)}$; π^2 appears in radial integrals on $V_{(1/2,1/2)}$ spinor tower as Hua kernel exponent contribution. - Exponent $6 = C_2$: substrate primary (quadratic Casimir of $B_2 = 6$).

This is a THREE-substrate-primary identification: numerator = K-type integer, denominator = Bergman π -factor, exponent = Casimir primary. All three pieces have substrate-mechanism candidates.

2. Bergman matrix element framework (mechanism candidate)

Per Tier 0 v0.1.6 + Hardy decomposition + Faraut-Korányi: matrix elements of operators between K-types V_λ on $H^2(D_{IV}^5)$ are computed by Bergman kernel integrals on the Shilov boundary.

For mass-related observables on the lepton sector ($V_{(1/2,1/2)}$ ground):

$$**m_\lambda^2 \propto \langle V_\lambda | M | V_\lambda \rangle = \int \partial_S K(\lambda, \lambda; \omega) \cdot M(\omega) \cdot d\mu_{FK}(\omega)**$$

where M is the mass operator (candidate: H_B per Tier 0 v0.1) and $K(\lambda, \lambda; \omega)$ is the kernel matrix element on the Shilov boundary at point ω .

For the lepton sector specifically: - Electron K-type: $V_{(1/2,1/2)}$ (Shilov primitive per Lane E candidate). $C_2 = 4$. Mass anchor. - Muon K-type: candidate identification at $V_{(1/2, 1/2)} +$ (excited radial mode N). $C_2 = 4 +$ (radial-tower contribution). - Tau K-type: $V_{(1/2, 1/2)} +$ (further excited radial mode). $C_2 = \dots$

Per Tier 0 v0.1.6 mass formula (R2): $m_\lambda \propto \sqrt{(C_2(\lambda) - C_2(\text{sector_ground}))}$.

But $(24/\pi^2)^6$ is NOT a simple $\sqrt{(C_2 \text{ difference})}$. It's a sixth-power structural form. This means the mass map is more involved than pure Casimir difference; the Bergman kernel matrix elements contribute the π^2 factor and the sixth-power structure.

3. The candidate Bergman-matrix-element derivation chain

Step 1 (substrate spinor tower): $V_{(1/2,1/2)}$ carries the spinor representation of $K = SO(5) \times SO(2)$. The radial-mode tower $V_{(1/2,1/2)}^{(n)}$ for $n = 0, 1, 2, \dots$ has Casimir levels $C_2^{(n)} = 4 + n(n + \rho_1 + \rho_2)$ where $\rho = (5/2, 3/2)$. So $C_2^{(0)} = 4$ (electron); $C_2^{(1)} = 4 + (1 + 4) = 9$ (muon candidate); $C_2^{(2)} = 4 + (2 + 6) = 12$ (tau candidate). [CANDIDATE — needs explicit verification.]

Step 2 (Bergman matrix elements on tower): matrix elements $\langle V^{(n)} | H_B | V^{(n)} \rangle$ involve Bergman kernel integrals over D_{IV}^5 at the spinor weight. By Faraut-Korányi Ch. XIII, the integral evaluates to a rational function of n times π^2 (the Bergman-volume contribution).

Step 3 (the closed form $(24/\pi^2)^6$): when you write out m_μ/m_e using the matrix elements from Step 2, the substrate-primary structure $(N_c, |W(B_2)|, C_2)$ emerges naturally, and the π^2 appears in the denominator from the Bergman volume factor. The sixth power = C_2 comes from the iterated structure of the radial tower (each radial mode contributes one factor; the muon = first excited radial mode requires $C_2 = 6$ iterations to reach observed mass ratio).

Step 4 (operator-level verification): explicit computation of $\langle V_{(1/2,1/2)}^{(1)} | H_B | V_{(1/2,1/2)}^{(1)} \rangle / \langle V_{(1/2,1/2)}^{(0)} | H_B | V_{(1/2,1/2)}^{(0)} \rangle$ using Mehler/Heckman-Opdam heat kernel on $H^2(D_{IV}^5)$. This is Elie's partial Mehler-kernel computation lane this afternoon (Toy 3659+ target per Keeper green-light).

4. What v0.2 commits to vs what stays OPEN

COMMITTED (v0.2 candidate): - $T_{190} = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$ is the substrate-decomposable reading. - The numerator integer (24) comes from K-type / Weyl-orbit structure (Saturday-confirmed). - The denominator π^2 comes from Bergman kernel matrix-element computation on $V_{(1/2,1/2)}$ spinor tower (FK Ch. XIII). - The exponent $6 = C_2$ comes from the radial-tower iteration structure (CANDIDATE; needs explicit verification).

OPEN (multi-week verification per Elie Mehler-kernel work): - Explicit Bergman matrix element on $V_{(1/2,1/2)}$ spinor tower gives the $(24/\pi^2)^6$ form (Elie Toy 3659+). - Mehler kernel partial computation today proves the matrix element framework is computable; multi-week extends to full closed form. - Casimir levels $C_2^{\{n\}}$ for excited spinor tower modes ($n=1$ muon, $n=2$ tau) need explicit calculation. - Per-sector ground subtraction per Tier 0 v0.1 R2: does the lepton-sector ground $V_{(1/2,1/2)}$ at $C_2 = 4$ give consistent mass anchor across leptons?

5. Connection to Tier 0 v0.1.6 boundary unification

Per v0.1.6: substrate's degrees of freedom live on Shilov $\partial_S D_{IV^5}$; bulk $H^2(D_{IV^5})$ is holomorphic extension.

L4 mass mechanism per this frame: - Mass operator M lives on the Shilov boundary (where the substrate writes commitments; mass is what gets recorded). - Bulk K-type V_λ has boundary value via Szegő projection; mass matrix element is computed boundary-side. - Bergman volume factor π^2 comes from the Hardy decomposition integral on Shilov $S^4 \times S^1/Z^2$. - The “iterated Casimir” structure (sixth power = C_2) comes from boundary K-type tower coupling.

So the L4 derivation operates at the BOUNDARY level (where mass is written), with the bulk K-type extension providing the substrate-primary integer ($N_c \cdot |W(B_2)|$) and the Casimir exponent (C_2). This is consistent with the Tier 0 v0.1.6 framework + the bulk-color v0.7 mechanism (which also operates boundary-side).

6. Generalization across SM masses (multi-week target)

If the $(24/\pi^2)^{C_2}$ form is verified for T190 m_μ/m_e , the same framework should reproduce: - T2003 m_τ/m_e : substrate reading $m_\tau/m_e = g^2 \cdot (2^{C_2} + g) = 49 \cdot 71 = 3479$ (Saturday). Candidate Bergman form: $(g \cdot (2^{C_2} + g) / \text{something})^{\text{rank?}}$. Needs work. - T187 $m_p/m_e = 6\pi^5$: candidate Bergman form involving N_c factor + π^{N_c} (Bergman volume on D_{IV^5}). - $m_\pi = \text{rank} \cdot N_{\max} \cdot m_e$ (Saturday): involves $N_{\max} = N_c^3 \cdot n_C + \text{rank}$. - $m_K = (g \cdot N_{\max} + \text{rank} \cdot n_C) \cdot m_e$ (Saturday): involves $N_{\max}$ additive combinations.

Each Saturday Tier 2 STRUCTURAL match should reduce to a Bergman matrix element form. v0.2 commits to T190 first; multi-week extends.

7. Honest scope + tier

RIGOROUS (this v0.2): - T190 numerator $24 = N_c \cdot |W(B_2)|$ (integer arithmetic, Saturday-confirmed). - T190 denominator π^2 has substrate-natural Bergman-volume interpretation (Faraut-Korányi standard). - Bergman matrix element framework for mass operators on $H^2(D_{IV^5})$ is well-defined (FK Ch. XIII).

CANDIDATE (v0.2's load-bearing): - $(24/\pi^2)^6 = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$ structural reading. - Spinor radial tower $V_{(1/2,1/2)}^{\{n\}}$ carries excited lepton mass spectrum. - Bergman matrix elements on tower give the closed-form structure.

FRAMEWORK (multi-week, Elie partial-Mehler-today + multi-week-full): - Explicit Mehler/Heckman-Opdam heat kernel computation for $H^2(D_{IV^5})$ (Elie green-lit afternoon work). - Explicit matrix element $\langle V^{\{n\}} | H_B | V^{\{n\}} \rangle$ at $n = 0, 1, 2$ (electron, muon, tau candidates). - Closed-form derivation of $(24/\pi^2)^{C_2}$ from Bergman integrals. - Promotion: SEARCH-FIT $\rightarrow$ DERIVATION when explicit computation matches.

Cal #27 / #182 / #99 discipline: v0.2 commits to a structural reading that is substrate-primary in three places (integer, π -factor, exponent). Cal #27 brake check: this feels like a clean decomposition. The brake holds because the decomposition is FRAMEWORK + CANDIDATE — the actual derivation requires Elie's Bergman matrix element computation. Until that lands, this is structural reading, not derivation. NOT promoted to RIGOROUS without explicit Bergman integral verification.

Two-Tier discipline (Elie Toy 3648): the integer part ($24 = N_c \cdot |W(B_2)|$) is Tier 1 EXACT. The matrix-element form $((24/\pi^2)^6)$ is Tier 2 STRUCTURAL at the natural precision floor (3.4×10^{-5} for T190). The mechanism-derivation target is to elevate from Tier 2 STRUCTURAL to RIGOROUS via the explicit Bergman computation.

8. Tension with v0.1 Section 3.2 (transparency)

Tier 0 v0.1 Section 3.2 listed $V_{(1/2,1/2)}$ with $C_2 = 4$ (spinor Casimir). v0.2 here uses the same $V_{(1/2,1/2)}$ as lepton sector ground (electron K-type candidate).

Apparent tension: per v0.1 mass $= \sqrt{C_2}$, electron mass would be $\sqrt{4} = 2$ (some unit). Per v0.2 here, T190 derivation uses radial tower $V_{(1/2,1/2)}^{\{n\}}$, with $n=0$ electron, $n=1$ muon, $n=2$ tau, etc. The “ground” for the spinor tower is the $n=0$ mode $V_{(1/2,1/2)}^{\{0\}}$, with the SECTOR GROUND for leptons.

Per Tier 0 v0.1 R2 (per-sector ground subtraction): $m_{\text{lepton}} = \sqrt{(C_2(V_{(1/2,1/2)}^{\{n\}}) - C_2(V_{(1/2,1/2)}^{\{0\}}))} = \sqrt{(\text{radial-tower contribution})}$.

For $n=0$ (electron): $m_e = \sqrt{0} = 0$ according to R2. This is wrong (electron has nonzero mass).

So R2 needs revision: the lepton sector ground is NOT $V_{(1/2,1/2)}^{\{0\}}$ itself; it's the GLOBAL ground $V_{\{(0,0)\}}$ (vacuum) per a different subtraction. Then $m_e = \sqrt{(C_2(V_{(1/2,1/2)}^{\{0\}}) - C_2(V_{\{(0,0)\}}))} = \sqrt{4} = 2$ (in substrate units), giving m_e its substrate mass anchor.

Revised reading: every K-type's mass is $\sqrt{C_2(\lambda)}$ (using vacuum $V_{\{(0,0)\}}$ as global ground). The “per-sector ground subtraction” of v0.1 R2 is not needed for absolute mass; it's needed for relative gauge-invariant comparisons.

This is a v0.1 R2 refinement; v0.2 flags it for Session 2 with Keeper.

9. Routing

→ Casey: L4 v0.2 commits to $(24/\pi^2)^6 = (N_c \cdot |W(B_2)|/\pi^2)^{C_2}$ structural decomposition reading for T190. First F2 test of Tier 0 framework. Elie green-lit partial Mehler kernel afternoon = first computational step in the multi-week explicit derivation. v0.1 R2 (per-sector ground subtraction) refined: vacuum $V_{\{(0,0)\}}$ is the global ground; per-sector subtraction is for gauge-invariant comparisons not absolute mass. Flagged for Session 2.

→ Elie: your partial Mehler kernel work today (Toy 3659+, ~25 K-type catalog data) directly tests the Bergman matrix element framework for $V_{(1/2,1/2)}$ spinor tower. Specific target: compute $\langle V_{(1/2,1/2)}^{\{0\}} | H_B | V_{(1/2,1/2)}^{\{0\}} \rangle$ via Faraut-Korányi Ch. XIII integral. If you can extract the π^2 factor structurally, that's the first F2 verification.

→ Keeper: L4 v0.2 stands ready for K-audit (“K-audit-L4-v0.2” item on your list). Session 2 with you should pin: (a) refined per-sector vs vacuum-ground formulation; (b) Bergman matrix element framework for mass operators; (c) explicit T190 derivation when Elie computes.

→ Grace: catalog T190 substrate-decomposition reading at CANDIDATE; cross-reference to Saturday's $24 = N_c \cdot |W(B_2)|$ reading + Tier 0 v0.1.6 boundary unification + Bergman matrix element framework.

→ Cal: cold-read welcome for “K-audit-L4-v0.2” item; specific Cal #27 concern: the $(24/\pi^2)^{C_2}$ structural reading feels too neat (all three substrate-primary). Please brake if the decomposition is post-hoc fitting rather than structural derivation. The actual derivation lives in Elie's Bergman matrix element computation; until that lands, this is structural reading not mechanism.

→ me: continuing to Lane E (Dictionary candidates) next per Casey afternoon plan; then P1 v0.7 cleanup; then EOD.

— Lyra, L4 v0.2 kernel-integral mass mechanism (first F2 test of Tier 0 framework). T190 $m_\mu/m_e = (24/\pi^2)^6$ reads substrate-decomposable as $(N_c \cdot |W(B_2)|/\pi^2)^{C_2}$: three substrate-primary identifications (integer numerator from K-type/Weyl-orbit, π^2 denominator from Bergman volume, C_2 exponent from substrate Casimir). Explicit derivation via Elie's Mehler kernel matrix element computation on $V_{(1/2,1/2)}$ spinor radial tower is the multi-week verification. v0.1 R2 per-sector subtraction refined: vacuum $V_{\{(0,0)\}}$ is global ground; per-sector for gauge-invariant comparisons only. F2 test FRAMEWORK ready; Elie afternoon Mehler-partial gives first computational step.